

# Performance Monitoring and Illumination Redesign of Academic Buildings With Feasibility Assessment of a Hybrid Solar Power System: A Case Study of Kathmandu University

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**ABSTRACT** Interior lighting is one of the dominant consumers of electrical energy in institutional buildings, commonly accounting for more than 30% of total electrical consumption. This paper reports a combined energy-audit and lighting-redesign study, together with a feasibility assessment of a hybrid solar power system, carried out at Blocks 2, 8 and 9 of Kathmandu University. Watt-hour load profiles were compiled for each block by on-site survey across working and non-working hours and across summer and winter seasons. On-site illuminance was measured with a digital lux meter and cross-checked against DIALux simulation, revealing that the fluorescent- and CFL-based lighting in Block 8 fell well short of recommended levels, whereas the LED lighting of Block 9 was closer to, though still below, the recommended illuminance. Replacement of the existing fixtures with LED luminaires was evaluated for Blocks 2 and 8, yielding an energy-consumption reduction of approximately 60% and simple payback periods of about 4 years and 12 years, respectively. A hybrid solar photovoltaic system was sized for the combined load of Blocks 8 and 9, requiring approximately 226 modules of 250 W (56.5 kWp), a 16-unit, 200 Ah battery bank, a single 292 A charge controller and a 100 kW hybrid inverter. The existing loop-in wiring scheme of Block 9 was also documented and analyzed for cable and protective-device sizing. The results support a priority order of solar photovoltaic supply, grid supply, battery storage and diesel back-up, and recommend LED retrofit and structured solar deployment as cost-effective measures for reducing the University's energy footprint.

**INDEX TERMS** Cable sizing, DIALux, hybrid solar power system, illuminance, LED retrofit, load profiling, loop-in wiring, payback period, photovoltaic sizing.

## I. INTRODUCTION

A Building's interior lighting system is both a dominant consumer of electrical energy and a major source of internal heat gain. In commercial and educational institutions, lighting typically accounts for more than 30% of total electrical energy consumption, yet a substantial share of this expenditure can be avoided through improved design and more efficient light sources [4]. Hybrid power systems combine two or more generation or storage technologies, most commonly a photovoltaic (PV) array with battery storage, so that solar energy can be stored during the day and drawn upon during a grid outage; such systems can serve as an

effective uninterruptible supply while reducing dependence on the utility grid [7], [12].

The work reported here was carried out on Blocks 2, 8 and 9 of Kathmandu University (KU) and pursued four related objectives: (i) quantify the electrical load carried by each block, (ii) evaluate the adequacy of the installed lighting systems against recommended illuminance levels, (iii) size a hybrid solar PV system capable of supplying part of the combined load of Blocks 8 and 9, and (iv) document and analyze the electrical wiring scheme of Block 9 to understand its cable sizing, protective-device ratings and fault characteristics.

## A. METHODOLOGY OVERVIEW

The study followed a four-stage methodology. First, the wattage consumption of every load (lighting, fans, personal computers, heaters, and other appliances) was surveyed room-by-room in Blocks 2 and 9, separated into working- and non-working-hour use, six days per week, and further split by summer and winter seasonal use. Second, illuminance produced by the installed dome, panel and tube-light fixtures was measured experimentally with a digital lux meter, compared with recommended illuminance levels, and cross-verified in DIALux. Third, the load data were used to size the PV array, battery bank, charge controller and inverter of a hybrid solar system, informed by a site visit to an in-progress solar installation at the Kathmandu University School of Medical Sciences (KUSMS), Dhulikhel. Fourth, under the guidance of the project supervisors, the loop-in wiring layout of Block 9 was studied to identify cable sizes, breaker ratings and possible fault mechanisms.

The remainder of this paper is organized as follows. Section II reviews the relevant photometric, electrical-protection and software background. Section III describes the system and measurement methodology in more detail. Section IV presents the measured load profiles. Section V presents the illuminance assessment and LED replacement analysis. Section VI documents the electrical wiring scheme of Block 9. Section VII presents the sizing of the hybrid solar PV system. Section VIII presents the overall results and discussion, and Section IX concludes the paper.

## II. BACKGROUND

### A. PHOTOMETRIC FUNDAMENTALS

The following photometric quantities were used throughout the illuminance assessment [2]. Luminance (or brightness),  $L$ , is the luminous intensity per unit projected area of a surface in a given direction,

$$L = \frac{I}{A \cos \theta} \quad [\text{cd/m}^2], \quad (1)$$

where  $I$  is the luminous intensity (cd) of a surface of area  $A$  seen at angle  $\theta$  to the normal. Luminous intensity,  $I = F/\omega$  (lm/sr), is the luminous flux  $F$  emitted per unit solid angle  $\omega$ . Luminous flux is the quantity of light energy emitted per second from a source, measured in lumens. Illuminance,  $E$ , is the luminous flux received per unit area of a surface,

$$E = \frac{F}{A} \quad [\text{lm/m}^2 \text{ or lux}]. \quad (2)$$

The lumen is the unit of luminous flux; a source of one candela emits a total of  $4\pi$  lumens. Candle power is the number of lumens emitted per unit solid angle in a given direction. The spectral power distribution (SPD) describes the energy of a light source as a function of wavelength, while color temperature, expressed in kelvin, describes whether a light source appears visually warm (yellowish, low K) or cool (bluish, high K).

Recommended illuminance depends on the activity performed in a space; for example, easy office work and classrooms are typically recommended around 250 lux, while normal office work, study and laboratory spaces are recommended around 500 lux [23], [24].

### B. HYBRID SOLAR POWER SYSTEMS

A hybrid solar power system couples a PV array with battery storage so that solar energy can be used during peak hours, stored for later use, and drawn upon during a grid outage, providing an uninterruptible supply function while reducing grid demand [12]. The principal advantages are the ability to store cheaper off-peak or solar energy, continued power availability during a blackout, and reduced dependence on a single source. The principal disadvantages are the higher capital cost associated with batteries, the larger installation footprint, the finite (7–15 year) service life of batteries, and a capacity limit on simultaneously running appliances that depends on the inverter rating.

### C. CABLE SIZING AND PROTECTIVE DEVICES

Cable selection is governed primarily by the current-carrying requirement of the circuit and by the operating voltage, since the maximum voltage a cable can safely carry depends on conductor size, insulation and construction [3]. A miniature circuit breaker (MCB) is generally used for lighting and small domestic/branch circuits below about 100 A, with a fixed trip characteristic, while a moulded case circuit breaker (MCCB) is rated from about 10–2500 A, combines thermal overload and magnetic short-circuit protection, and is typically used for commercial and industrial distribution boards carrying more than 100 A [11].

### D. SOFTWARE AND INSTRUMENTATION

Three tools supported the study. AutoCAD Electrical was used for schematic and panel-layout drafting of the wiring diagrams. DIALux was used to simulate and visualize illuminance distribution for single rooms and whole floors, including import of AutoCAD architectural plans [21]. A digital illuminance (lux) meter was used for on-site measurement of illuminance, expressed in lux (lm/m<sup>2</sup>).

## III. SYSTEM AND METHODOLOGY

### A. SITE DESCRIPTION

The study covers three academic blocks at Kathmandu University: Block 2 (a multi-floor academic/administrative building), Block 8 (an academic block with CFL-based lighting reported by occupants as uneven and insufficient), and Block 9 (an academic block with a predominantly LED and dome/panel-light installation, and whose electrical wiring scheme was also analyzed in detail).

### B. LOAD DATA COLLECTION

Every room of Blocks 2 and 9 was surveyed to identify the type, quantity, and wattage of each connected load (lights,

fans, personal computers, printers, heaters, boilers and other appliances). Because occupancy and usage vary with time of day and season, the survey distinguished (i) working hours from non-working hours, (ii) a six-day working week (with the lowest usage occurring on Saturday), and (iii) summer loads from winter loads, the latter including seasonal heating appliances not used in summer. For each appliance, the average watt-hours per day was computed as

$$\overline{Wh}_{\text{day}} = \frac{N \times P \times h \times d}{7}, \quad (3)$$

where  $N$  is the appliance count,  $P$  the rated wattage,  $h$  the hours of use per day and  $d$  the days of use per week.

### C. ILLUMINANCE MEASUREMENT

For each room of interest, the floor area was partitioned into a grid of sub-areas (for example, Room 308 of Block 9 was partitioned into 40 sub-areas) and illuminance was recorded at each sub-area with a digital lux meter; the arithmetic mean of the sub-area readings gave the room's average illuminance. The same procedure was repeated for several representative rooms of Block 8 to compare CFL- and LED-based illumination, and the resulting illuminance levels were cross-checked against a DIALux simulation of each space.

### D. HYBRID SOLAR PV SIZING PROCEDURE

The combined corrected daily energy demand of Blocks 8 and 9 was used to size, in sequence, the required PV array wattage, the number of panels in series and parallel, the battery bank capacity and unit count, the charge-controller current rating, and the inverter power rating, following the standard load-based PV sizing procedure [15]. The procedure is described quantitatively in Section VII.

### E. STUDY OF THE BLOCK 9 WIRING SCHEME

Under the guidance of the project supervisors, the existing loop-in wiring layout of Block 9 was traced from the main distribution board down to the individual room MCBs, in order to record conductor cross-sections, breaker ratings, and the branch topology, and to identify possible fault mechanisms and mitigations. This analysis is presented in Section VI.

## IV. LOAD PROFILE ANALYSIS

Tables 1–3 summarize the measured load profiles. Block 2 draws an average of 167.89 kWh/day, with a pronounced winter increase driven by seasonal heating loads on the ground and first floors. Block 9 draws an average of 105.23 kWh/day, with the top (third) floor accounting for the largest share of consumption in both wings. Block 8's lighting load alone reaches 138.89 kWh/day (summer), concentrated on the third and fourth floors, which is consistent with the poor and uneven illumination reported by occupants of that block (Section V).

TABLE 1. Load Profile of Block 2

| Floor           | Summer (Wh/day) | Winter (Wh/day) |
|-----------------|-----------------|-----------------|
| Basement        | 2 376.34        | 2 347.20        |
| Ground          | 21 427.80       | 92 827.80       |
| First           | 84 711.08       | 122 430.85      |
| Second          | 112 585.71      | 103 390.28      |
| Third           | 19 874.80       | 12 025.25       |
| Total           | 240 975.33      | 333 021.38      |
| Demand factor   | 0.50            | 0.65            |
| Total (kWh/day) | 120.49          | 216.46          |

Average total load: 167.89 kWh/day.

TABLE 2. Load Profile of Block 9

| Floor              | Summer (Wh/day) | Winter (Wh/day) |
|--------------------|-----------------|-----------------|
| First              | 9 195.43        | 2 900.57        |
| Second             | 27 366.86       | 37 858.28       |
| Third (North Wing) | 58 230.86       | 30 000.00       |
| Third (South Wing) | 69 143.31       | 68 587.88       |
| Fourth             | 26 670.17       | 37 819.88       |
| Total              | 190 606.63      | 177 166.61      |
| Demand factor      | 0.50            | 0.65            |
| Total (kWh/day)    | 95.30           | 115.16          |

Average total load: 105.23 kWh/day.

TABLE 3. Load Profile of Block 8 (Lighting)

| Floor           | Summer (Wh/day) |
|-----------------|-----------------|
| First           | 3 676.94        |
| Second          | 65 863.03       |
| Third           | 78 659.40       |
| Fourth          | 105 704.54      |
| Fifth           | 23 335.64       |
| Others          | 537.43          |
| Total           | 277 776.98      |
| Demand factor   | 0.50            |
| Total (kWh/day) | 138.89          |

TABLE 4. Recommended Illuminance Levels

| Activity / space                        | Illuminance (lux) |
|-----------------------------------------|-------------------|
| Public areas with dark surroundings     | 20–50             |
| Simple orientation, short visits        | 50–100            |
| Occasional visual tasks                 | 100–150           |
| Warehouses, homes, theaters, archives   | 150               |
| Easy office work, classrooms            | 250               |
| Normal office work, study, laboratories | 500               |
| Supermarkets, mechanical workshops      | 750               |
| Detailed drawing / mechanical work      | 1000              |

## V. ILLUMINANCE ASSESSMENT AND LED REPLACEMENT ANALYSIS

### A. RECOMMENDED ILLUMINANCE LEVELS

Table 4 lists representative recommended illuminance levels used as the benchmark for this study [23].

### B. MEASURED ILLUMINANCE

Partitioning Room 308 (third floor, Block 9) into 40 sub-areas and averaging the lux-meter readings gave a mean

**TABLE 5.** Summary of Lighting Load Before and After LED Retrofit

|                                   | Block 2        | Block 8        |
|-----------------------------------|----------------|----------------|
| Existing total wattage (W)        | 10 102         | —              |
| LED-retrofit total wattage (W)    | 4 042          | —              |
| Existing energy (Wh/month)        | 1 212 240      | 917 400        |
| LED-retrofit energy (Wh/month)    | 485 040        | 589 320        |
| Existing monthly cost             | NRs. 13 334.64 | NRs. 10 091.40 |
| LED-retrofit monthly cost         | NRs. 5 335.55  | NRs. 6 482.52  |
| Installation cost of LED retrofit | NRs. 379 090   | NRs. 503 240   |

illuminance of **158.3 lux**, below the 250 lux recommended for classroom use. Comparative spot measurements in several rooms of Block 8 gave substantially lower and more uneven illuminance, with room averages ranging from approximately 18–79 lux across the sampled rooms, confirming that the existing fluorescent/CFL lighting of Block 8 falls well short of the recommended level and exhibits pronounced spatial unevenness. Occupants of Block 8 reported shadow effects, eye strain and discomfort consistent with these deficiencies, which in turn were judged likely to reduce productivity in tasks requiring visual precision.

### C. LED REPLACEMENT ANALYSIS

For an equal rated wattage, the LED fixtures installed in Block 9 produced measurably higher illuminance than the CFL and fluorescent fixtures of Block 8. Table 5 summarizes the wattage and cost implications of replacing the existing lighting of Blocks 2 and 8 with LED fixtures (down-lights, panel lights and dome LEDs sized to match the required illuminance in each space).

The reduction in lighting energy consumption from the LED retrofit is approximately 60% for both blocks. LED fixtures were preferred over the incumbent sources principally for their longer operating life, higher energy efficiency, environmental profile, mechanical durability, design flexibility, cold/hot-temperature tolerance, instant switching, and compatibility with low-voltage supply.

### VI. ELECTRICAL WIRING SCHEME OF BLOCK 9

Block 9 is wired using a loop-in system, in which lamps and switches are connected in parallel and the phase (line) conductor is looped from one outlet directly to the next, allowing each fitting to be controlled independently [17]. Line or phase conductors are looped only within a switchboard or accessory box, and never through a lamp or fan fitting. Fig. 1 illustrates the basic loop-in wiring principle as applied in the building.

Fig. 2 reproduces the distribution schematic of Block 9 as drafted in AutoCAD Electrical for this project. A 95 mm<sup>2</sup> feeder, rated for 150 A, supplies the main distribution board, which is not protected by an MCCB – identified in this study as a shock-protection gap at that point. From the main board, three 50 A MCCB-protected branches feed, respectively, the South Wing (via 6 mm<sup>2</sup> cable), the North Wing (via 4 mm<sup>2</sup> cable) and the Civil Engineering laboratories. Each branch in turn feeds a row of miniature circuit breakers (rated

**TABLE 6.** Combined Load Used for Solar PV Sizing (Blocks 8 and 9)

| Quantity                                | Value             |
|-----------------------------------------|-------------------|
| Total connected wattage (instantaneous) | 70 560.00 W       |
| Total energy demand                     | 248 033.27 Wh/day |
| Load correction factor                  | 0.15              |
| Corrected energy demand                 | 285 238.26 Wh/day |

C6/C16/C20/C25 at 240/415 V) supplying individual rooms and laboratories, identified in the diagram by room number.

Each of the three 6 mm<sup>2</sup> distribution-board feeders is rated for approximately 30 A continuous, with a practical overload margin to 40 A; the branch MCCBs are set to a nominal 50 A, so that current is intentionally limited below the theoretical rating to protect the cable. Because the 150 A main feeder never in practice carries the arithmetic sum of the three 50 A branch ratings simultaneously, balancing across the branches is not a practical concern under normal loading. From the North and South Wing boards, risers to the upper floors use two 4 mm<sup>2</sup> cables (protected by 25 A MCCBs) and one 6 mm<sup>2</sup> cable (protected by a 32 A MCCB) per floor, the latter reserved for computer-laboratory and other higher-power loads. In the Civil Engineering laboratories, a separate feeder distribution board is provided per room so that a fault can be isolated by disconnecting only the affected board. The principal advantages of the loop-in method are that it requires no joint boxes (reducing material cost), leaves no concealed joints beneath floors or in roof spaces, and simplifies fault location, since connections are made only at accessible outlets; its principal disadvantages are a greater total length of cable (and hence greater voltage drop and copper loss) and greater difficulty in looping switches and lamp holders during installation.

### VII. SIZING OF THE HYBRID SOLAR PV SYSTEM

The hybrid solar PV system was sized for the combined load of Blocks 8 and 9, following the procedure of Section III-D.

#### A. COMBINED LOAD AND CORRECTION

The combined connected wattage, the total daily watt-hour demand and the corrected daily demand (applying a 15% load-correction factor) are summarized in Table 6.

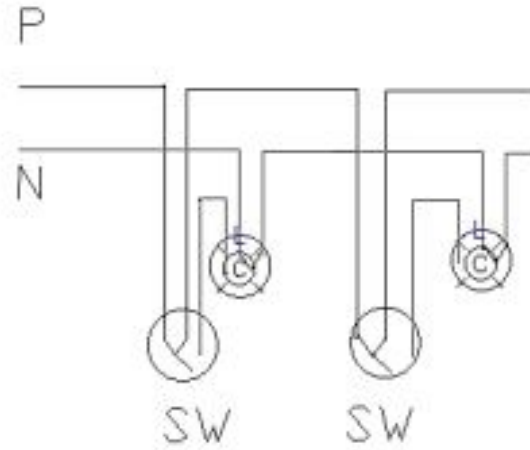
#### B. PV ARRAY SIZING

With an assumed 5 h/day of effective sun hours at the site, and using 250 W modules ( $V_{mp} = 96$  V,  $I_{max} = 2.60$  A), the required watt-peak rating and panel count follow from

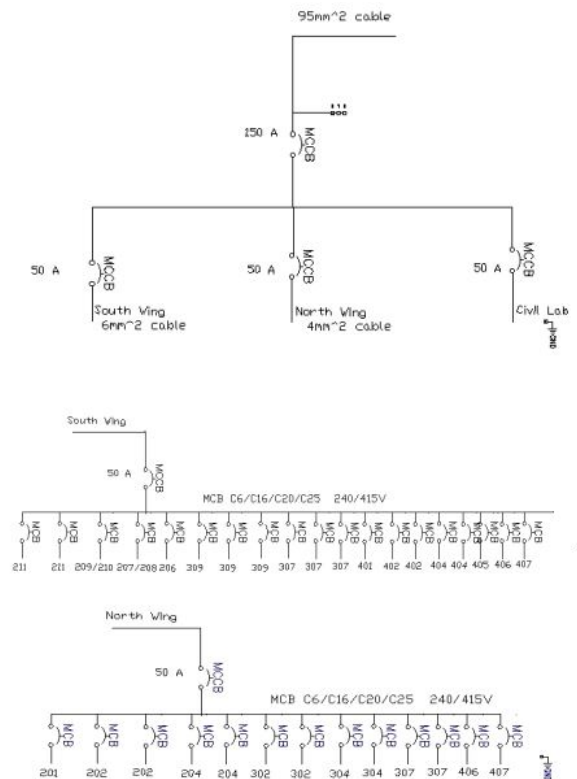
$$P_{\text{peak}} = \frac{Wh_{\text{corrected}}}{\text{sun hours} \times 1000}, \quad (4)$$

$$N_{\text{panels}} = \frac{P_{\text{peak}}}{P_{\text{module}}}. \quad (5)$$

This gives a required array rating of approximately 56.15 kWp, i.e. 225 modules (rounded), arranged as 2 panels in series and 113 strings in parallel at a system voltage of 192 V,



**FIGURE 1.** Loop-in wiring system used in Block 9 (as drawn in the original project schematics): the phase (P) and neutral (N) buses are looped in turn to each switch (SW) and lamp/ceiling-fan point.



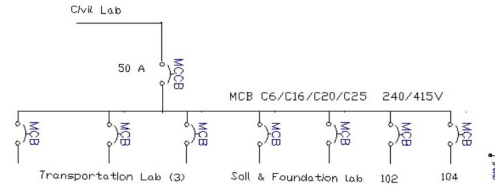
**FIGURE 2.** Electrical distribution schematic of Block 9: main 95 mm<sup>2</sup> feeder and 150 A MCCB, branching into the South Wing, North Wing and Civil Lab feeders (each protected by a 50 A MCCB), and the associated room-level MCB rows for the South and North Wings (as drafted in the original project schematics).

for a total of 226 modules after rounding for array symmetry (Table 7).

### C. BATTERY SIZING

Battery capacity was sized assuming that 10% of the total daily load is served from storage, using

$$C_B = \frac{E}{\text{DOD} \times \eta_B \times N_v}, \quad (6)$$



**FIGURE 3.** Room-level MCB row supplying the Civil Engineering laboratories from the Civil Lab branch of the Block 9 distribution board (as drafted in the original project schematics).

**TABLE 7.** PV Array Sizing Summary

| Quantity                                   | Value                 |
|--------------------------------------------|-----------------------|
| Effective sun hours                        | 5 h                   |
| Total watt-peak rating required            | 56.15 kW              |
| Module rating ( $P_{max}/I_{max}/V_{mp}$ ) | 250 W / 2.60 A / 96 V |
| Panels in series ( $N_s$ )                 | 2                     |
| Panels in parallel ( $N_p$ )               | 113                   |
| System voltage                             | 192 V                 |
| Total panels required                      | 226                   |

**TABLE 8.** Battery Bank Sizing

| Quantity                                        | Value        |
|-------------------------------------------------|--------------|
| Load to be served from storage (incl. 10% loss) | 30 881.09 Wh |
| Depth of discharge                              | 0.8          |
| Nominal battery voltage                         | 12 V         |
| Unit battery rating                             | 200 Ah       |
| Required bank capacity                          | 3 216.78 Ah  |
| Number of batteries required                    | 16           |

where  $E$  is the storage-served energy, DOD the depth of discharge,  $\eta_B$  the battery efficiency and  $N_v$  the nominal cell voltage (12 V). With a depth of discharge of 0.8 and 200 Ah, 12 V units, this yields a required capacity of 3 216.78 Ah and, hence, 16 batteries (Table 8).

#### D. CHARGE CONTROLLER AND INVERTER SIZING

The charge-controller operating voltage follows from the battery bank voltage (192 V), giving a required current rating of 292.43 A and hence a single charge controller of that rating (Table 9). Allowing a 30% margin above the 70.56 kW instantaneous connected load for future expansion, the required inverter rating is 91.73 kW; a 100 kW hybrid inverter (DC/AC coupled, with grid back-up capability) is therefore recommended.

#### E. FIELD VERIFICATION

Tilt-angle-versus-open-circuit-voltage measurements were carried out on the thin-film PV modules already installed on the roofs of Blocks 8 and 9. Voltage was observed to decline gradually as the panel tilt angle increased from normal incidence toward grazing incidence (e.g., from about 104 V at 0° to about 89 V near 90° in one measurement series), consistent with the expected cosine-law reduction in effective irradiance

**TABLE 9.** Charge Controller and Inverter Sizing

| Quantity                                   | Value    |
|--------------------------------------------|----------|
| Charge-controller operating voltage        | 192 V    |
| Required charge-controller current rating  | 292.43 A |
| Number of charge controllers               | 1        |
| Inverter operating voltage                 | 192 V DC |
| Connected load (inverter sizing basis)     | 70.56 kW |
| Required inverter rating (with 30% margin) | 91.73 kW |
| Recommended inverter rating                | 100 kW   |

**TABLE 10.** Simple Payback Period for LED Retrofit

|                          | Block 2    | Block 8     |
|--------------------------|------------|-------------|
| Installation cost (NRs.) | 379 090    | 503 240     |
| Monthly saving (NRs.)    | 7 999.09   | 3 608.88    |
| Payback period           | ≈ 47.4 mo. | ≈ 139.4 mo. |
| Payback period (years)   | ≈ 4        | ≈ 12        |

and confirming that the installed array responds as expected to changes in incidence angle.

## VIII. RESULTS AND DISCUSSION

### A. ENERGY SAVINGS

Replacing fluorescent and CFL lighting with LED fixtures in Blocks 2 and 8 reduces lighting energy consumption by approximately 60%, consistent with the wattage reductions reported in Table 5.

### B. PAYBACK PERIOD

The simple payback period for the LED retrofit, obtained by equating the cumulative cost of the installed LED system to the cumulative cost of continuing to operate the existing fixtures, is approximately **4 years** for Block 2 and approximately **12 years** for Block 8 (Table 10). The shorter payback for Block 2 reflects its comparatively lower LED installation cost relative to its baseline energy expenditure, whereas Block 8's larger and more extensive retrofit carries a correspondingly longer payback horizon.

### C. RECOMMENDED SOLAR POWER COMPONENTS

Table 11 summarizes the recommended ratings of the hybrid solar power system components sized in Section VII. The sizing supports a supply-priority order of (1) solar PV array,

**TABLE 11.** Summary of Recommended Solar Power Component Ratings

| Component         | Unit rating  | Quantity       |
|-------------------|--------------|----------------|
| Solar PV module   | 250 W        | 226 (56.5 kWp) |
| Battery           | 200 Ah, 12 V | 16 (3 200 Ah)  |
| Charge controller | 292.43 A     | 1              |
| Inverter          | 100 kW       | 1              |

**TABLE 12.** Illuminance Achieved After Redesign (DIALux evo)

| Area                        | Illuminance (lux) |
|-----------------------------|-------------------|
| Classrooms and laboratories | 250–300           |
| Staff rooms                 | 60–200            |
| Corridor areas              | 60–150            |
| Entrance areas              | 60–150            |
| Washrooms                   | 20–70             |
| Stores                      | 3–7               |
| Ladder/stair areas          | 5–25              |
| Vice-Chancellor's room      | 400               |
| Conference room             | 400               |

(2) Nepal Electricity Authority (NEA) grid supply, (3) battery storage, and (4) diesel generator back-up.

#### D. DIALUX-VERIFIED ILLUMINANCE

Following the LED-fixture redesign, DIALux simulation confirmed illuminance within the ranges recommended for each space category (Table 12), correcting the deficiencies identified for Block 8 in Section V.

#### E. LIMITATIONS

The reported sizing of the solar power components should be treated as a preliminary, load-based recommendation rather than a final engineering design: it does not yet incorporate a full protection coordination study, shading analysis, or detailed structural loading assessment of the roof-mounted array, and the MCCB ratings identified on the Block 9 distribution boards were, at the time of the survey, set by assumption rather than by calculation. Full floor-by-floor load registers and floor-wise loop-in wiring diagrams are provided in the accompanying appendix.

#### IX. CONCLUSION AND RECOMMENDATIONS

This study analyzed and quantified the electrical and lighting energy-saving potential of Blocks 2, 8 and 9 of Kathmandu University. Interior lighting was found to account for a major share of energy consumption in these blocks, and the poor and uneven illuminance delivered by the existing fluorescent/CFL lighting of Block 8 was identified as a likely source of visual discomfort and reduced productivity. Replacing the existing fluorescent and CFL fixtures with LED luminaires in Blocks 2 and 8 reduces lighting energy consumption by approximately 60%, with simple payback periods of about 4 and 12 years, respectively; despite the larger upfront cost of the LED modules and of the proposed solar PV system, both investments are recoverable within a reasonably short period given prevailing NEA tariffs. Documentation of the loop-in

wiring scheme of Block 9 further identified a specific protection gap (the absence of an MCCB at the main distribution board) and clarified the existing cable and breaker sizing. The concerned University authorities are recommended to prioritize (1) LED retrofit of Blocks 2 and 8, (2) implementation of the sized hybrid solar PV system with the supply-priority order of solar, grid, battery and diesel back-up, and (3) a follow-up protection-coordination study of the Block 9 distribution boards.

#### APPENDIX A FLOOR-WISE LOOP-IN WIRING DIAGRAMS OF BLOCK 9

Figs. reproduce the floor-by-floor loop-in wiring arrangement of Block 9, as drafted for this project, showing the lamp, ceiling-fan and switch points wired from each room's distribution point on Floors 1–4.

#### APPENDIX B DIALUX ILLUMINANCE AND LUMINANCE VERIFICATION PLOTS

Figs. present representative DIALux plots for one floor of each surveyed block, supporting the illuminance results summarized in Section V and Table 12. Full room-by-room load registers and the complete set of DIALux views for every floor of Blocks 2, 8 and 9 are available as supplementary material accompanying this manuscript.

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